



BACHELORS IN ARTIFICIAL INTELLIGENCE (BS AI)

Structure of BSAI Programs Total Credit Hours: 136 Total Courses: 44 Semesters: 8 Duration: 4 Years	Category: Courses (Credit Hours) General Education: 12 (30) Major courses: 18 (64) + 7 (21) Allied/interdisciplinary courses: 5 (15) Internship/field experience: N/A Capstone project: 2 (6) Total: 44 (136)
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SCHEME OF STUDIES

S #	COURSE CODES	CLASS	COURSE TITLE	CREDIT HOURS	PRE-REQ	CO-REQ
SEMESTER 1						
1	CMC111	CORE	PROGRAMMING FUNDAMENTALS	3 + 1		PROGRAMMING FUNDAMENTALS LAB
2	GER111	GER	APPLICATION OF INFORMATION & COMMUNICATION TECHNOLOGIES	2 + 1		APP. OF INFO.& COMM.TECH. LAB
3	GER151	GER	NATURAL SCIENCE (APPLIED PHYSICS)	2 + 1		APPLIED PHYSICS LAB
4	GER131	GER	CALCULUS AND ANALYTIC GEOMETRY	3 + 0		
5	GER121	GER	FUNCTIONAL ENGLISH	3 + 0		
6	GER141	GER	ISLAMIC STUDIES	2 + 0		
				18(15+3)		
SEMESTER 2						
1	CMC112	CORE	OBJECT ORIENTED PROGRAMMING	3 + 1	CMC111	OBJECT ORIENTED PROGRAMMING LAB
2	GER132	GER	DISCRETE STRUCTURES	3 + 0	GER131	
3	CMC121	CORE	DIGITAL LOGIC DESIGN	2 + 1	GER151	DIGITAL LOGIC DESIGN LAB
4	GER142	GER	IDEOLOGY AND CONSTITUTION OF PAKISTAN	2 + 0	-	
5	GER122	GER	EXPOSITORY WRITING	3 + 0	GER121	
6	MTE111	MATHS	MULTIVARIABLE CALCULUS	3 + 0	GER131	
				18 (16-2)		
SEMESTER 3						
1	CMC251	CORE	DATA STRUCTURES	3 + 1	CMC112	DATA STRUCTURES LAB
2	CMC222	CORE	COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE	3 + 1	CMC121	COMPUTER ORG. & ASSEMBLY LAN. LAB
3	AIC211	DOMAIN CORE	ARTIFICIAL INTELLIGENCE	2+ 1	CMC112	ARTIFICIAL INTELLIGENCE LAB
4	CMC261	CORE	COMPUTER NETWORKS	3 + 1	CMC121	COMPUTER NETWORKS LAB
5	MTE212	MATHS	PROBABILITY & STATISTICS	3 + 0	MTE111	
				18 (15-3)		
SEMESTER 4						
1	CMC241	CORE	OPERATING SYSTEMS	3 + 1	CMC251	OPERATING SYSTEMS LAB

2	AIC212	DOMAIN CORE	PROGRAMMING FOR AI	2 + 1	AIC211	ARTIFICIAL INTELLIGENCE LAB
3	MTE213	MATH	LINEAR ALGEBRA	3 + 0	MTE111	
4	CMC252	CORE	ANALYSIS OF ALGORITHMS	3 + 0	CMC251	
5	MTE221	GER	TECHNICAL & BUSINESS WRITING	3 + 0	GER122	
6	GERXXX	GER	SOCIAL SCIENCES (ELECTIVE I)	2 + 0	GER122	
				18 (16-2)		
SEMESTER 5						
1	CMC371	CORE	SOFTWARE ENGINEERING	3 + 0	CMC252	
2	CMC362	CORE	INFORMATION SECURITY	3 + 1	CMC261	INFORMATION SECURITY LAB
3	CMC331	CORE	DATABASE SYSTEMS	3 + 1	CMC241	DATABASE SYSTEMS LAB
4	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 1	3 + 0	-	
5	AIC221	DOMAIN CORE	INTRODUCTION TO MACHINE LEARNING	2 + 1	AIC211	INTRODUCTION TO MACHINE LEARNING LAB
				17(14+3)		
SEMESTER 6						
1	AIC322	DOMAIN CORE	ARTIFICIAL NEURAL NETWORKS & DEEP LEARNING	3 + 1	AIC221	ANN & DEEP LEARNING LAB
2	AIE423	DOMAIN CORE	COMPUTER VISION	2 + 1	AIC221/ AIC211	COMPUTER VISION LAB
3	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 2	2 + 1	-	
4	ESCXXX	GER	SOCIAL SCIENCE II	3 + 0	-	
5	AIC331	DOMAIN CORE	KNOWLEDGE REPRESENTATION & REASONING	2 + 1	CMC331	KNOWLEDGE REP. & REASONING LAB
				16(12+4)		
SEMESTER 7						
1	CMC491	CORE	FINAL YEAR PROJECT - I	0 + 3	CMC371 / AIC322	
2	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 3	2 + 1	-	
3	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 4	2 + 1	-	
4	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 5	2 + 1	-	
5	CMC442	CORE	PARALLEL & DISTRIBUTED COMPUTING	3 + 1	CMC241	PAR.& DISTRIBUTED COMPUTING LAB
6	GER462	GER	TECHNOPRENEURSHIP	2 + 0		
				18(11-7)		
SEMESTER 8						
1	CMC492	CORE	FINAL YEAR PROJECT - II	0 + 3	CMC491	
2	GER463	GER	PROFESSIONAL PRACTICES	2 + 0		
3	GER443	GER	CIVICS AND COMMUNITY ENGAGEMENT	2 + 0		
4	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 6	2 + 1	-	
5	CSEXXX	DOMAIN ELEC	DOMAIN ELECTIVE 7	2 + 1	-	
				13(8-5)		
			TOTAL CREDIT HOURS	136		

****Campus will decide to offer a course.**